OPERATOR HANDBOOK

N0JCG Winlink Email Server

Installation, client setup, webmail operation, and troubleshooting

A complete guide to the Raspberry Pi client appliance, Winlink mailbox access, DigiRig Mobile integration, and safe Packet RMS operation.

Release	Publication
0.1.4	August 2026
PRODUCT ROLE Client-side Winlink email	AUDIENCE Operators and mailbox users

At a glance

N0JCG Winlink Email Server places the operator console, Winlink Client mailbox, local drafts and queues, and the webmail experience on one Raspberry Pi appliance. The external radio and Packet RMS gateway provide the radio transport.

THE OPERATING MODEL

Browser -> N0JCG webmail and operator console -> N0JCG management layer -> Winlink Client mailbox and transport -> DigiRig Mobile -> radio -> external Packet RMS gateway.

Surface	Use it for	Address
Operator console	Configuration, status, diagnostics, and safety controls	http://PI-IP/ui/
User webmail	Inbox, compose, drafts, folders, signature, and queue	http://PI-IP/webmail/
Deployment helper	Install and upgrade from MSYS2 Bash	./deploy/push_to_pi.sh

Capability boundary

- The appliance can be installed and configured before the radio is connected.
- USB detection is evidence that the operating system saw a device; it is not proof of audio, PTT, packet, or RF operation.
- Transmission stays disabled until the operator explicitly configures and verifies the radio path.

1. Getting started

Required hardware

- Raspberry Pi 4 with 64-bit Debian/Raspberry Pi OS and reliable power
- microSD card with the operating system installed
- DigiRig Mobile audio/PTT interface and USB cable
- Compatible amateur radio with packet/data audio and PTT connections
- Radio-specific interface cables
- Ethernet or Wi-Fi network connection
- A workstation on the same network running MSYS2 Bash, Git, OpenSSH, and sshpass

Helpful accessories

- Powered USB hub for multiple peripherals
- Keyboard and display for recovery
- Labeled cables
- UPS or clean 5 V power for unattended use

PRE-RADIO SETUP

Software, branding, accounts, and webmail can be prepared before the radio is connected. Complete the software setup first, then bring up the radio as a controlled receive-first procedure.

2. Install from MSYS2 Bash

The official deployment helper copies the complete application to the Pi, runs the installer, and verifies that the webmail service is active.

Prepare the workstation

1. Open MSYS2 UCRT64 or MSYS2 MSYS Bash.
2. Install the deployment tools with: pacman -S --needed git openssh sshpass
3. Clone the repository: git clone https://github.com/bakstaaj/N0JCG-WINLINK-EMAIL-SERVER.git
4. Enter the repository: cd N0JCG-WINLINK-EMAIL-SERVER

Run the guided deployment

```
./deploy/push_to_pi.sh
```

The helper prompts for the Raspberry Pi IP address, SSH username, and SSH password. The default address is 192.168.68.149 and the default SSH username is pi, but these values are intended to be changed for another installation.

Operator authentication

During the initial install, the installer asks for an Nginx operator-console username, password, and password confirmation. Existing operator authentication is preserved during normal upgrades.

Wi-Fi, hotspot fallback, and USB gadget

To configure connectivity during deployment, add --configure-connectivity. The helper configures the secured N0JCG-WES hotspot and confirms its password. The Pi activates the hotspot at 192.168.50.1 with leases from 192.168.50.100 through 192.168.50.200. The initial password is Password; change it during installation when prompted.

```
./deploy/push_to_pi.sh 192.168.68.149 pi --configure-connectivity
```

The Pi 4 USB Ethernet gadget is available at 192.168.60.1 through the USB-C power/data port. The blue USB host ports are not gadget ports.

RECONFIGURE OPERATOR AUTH

Use ./deploy/push_to_pi.sh PI-IP PI-USER --configure-operator-auth when the operator account must be intentionally changed.

Verify

1. Open http://PI-IP/ui/ from a workstation on the same network.
2. Open http://PI-IP/webmail/.
3. Confirm the deployment helper reports that the API was deployed and n0jcg-webmail.service is active.

3. First-time setup

Operator console

Sign in to the operator console with the credentials created by the installer. Review the host identity, service state, DigiRig/USB detection, Winlink Client readiness, radio profile, Packet settings, and diagnostics before connecting RF equipment.

Find a nearby Packet RMS gateway

The Nearby RMS gateways panel uses the official Winlink gateway status data. With Internet access, select Refresh gateway list to update the local cache. The cached list remains available in the field without Internet access. The panel shows the five closest Packet RMS channels in a scrollable list. Enter simulated latitude/longitude for testing, or select Use GPS after a USB GPS receiver and gpsd are available.

Double-click a gateway row to copy its RMS target and frequency into the Packet station profile, scroll to that profile, and focus the target field. Review the values before selecting Save radio profile. The selected target applies to the next Winlink client exchange; it does not change the external gateway.

Webmail account

Open the webmail address and select Sign in with Winlink. Enter the user's Winlink email address and secure-login password. The appliance validates the credentials against Winlink before opening the private local mailbox for that callsign. Webmail remains closed until Winlink CMS secure login is accepted; failed or timed-out authentication never opens a session or mailbox. After secure login succeeds, the session remains active for the browser session; refreshing the page does not sign the user out. Close the browser or select Log out to end the session.

PRIVATE MAILBOX MODEL

Mailbox access is isolated by callsign. A user cannot select another user's mailbox from a URL or browser control.

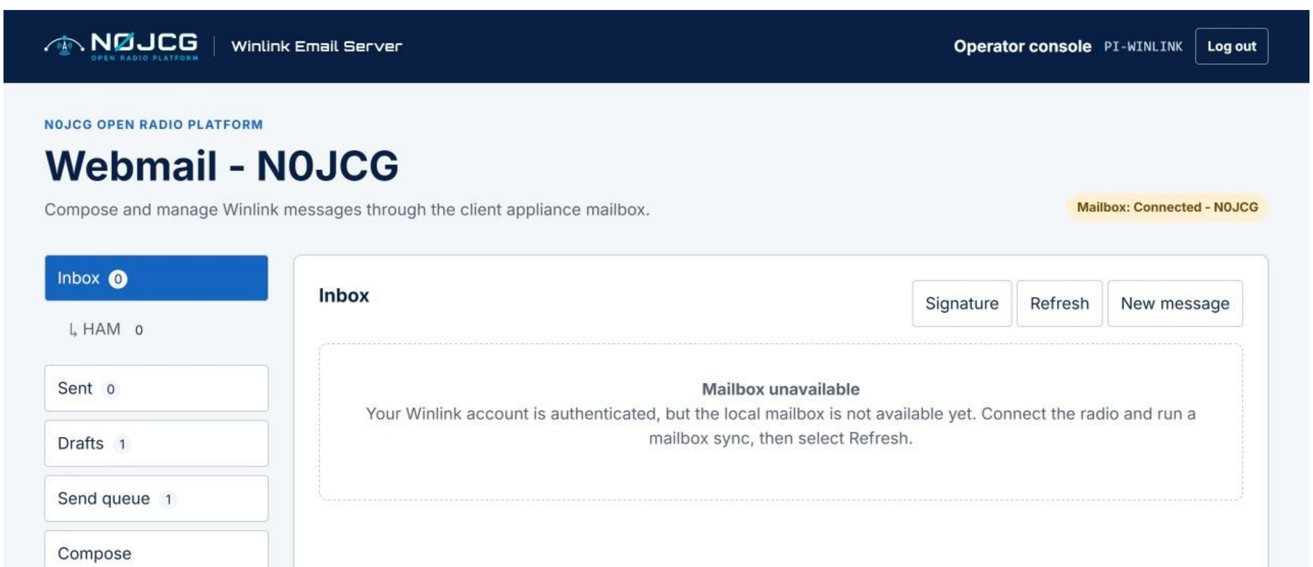

Figure 1. N0JCG webmail inbox and mailbox status on the client appliance.

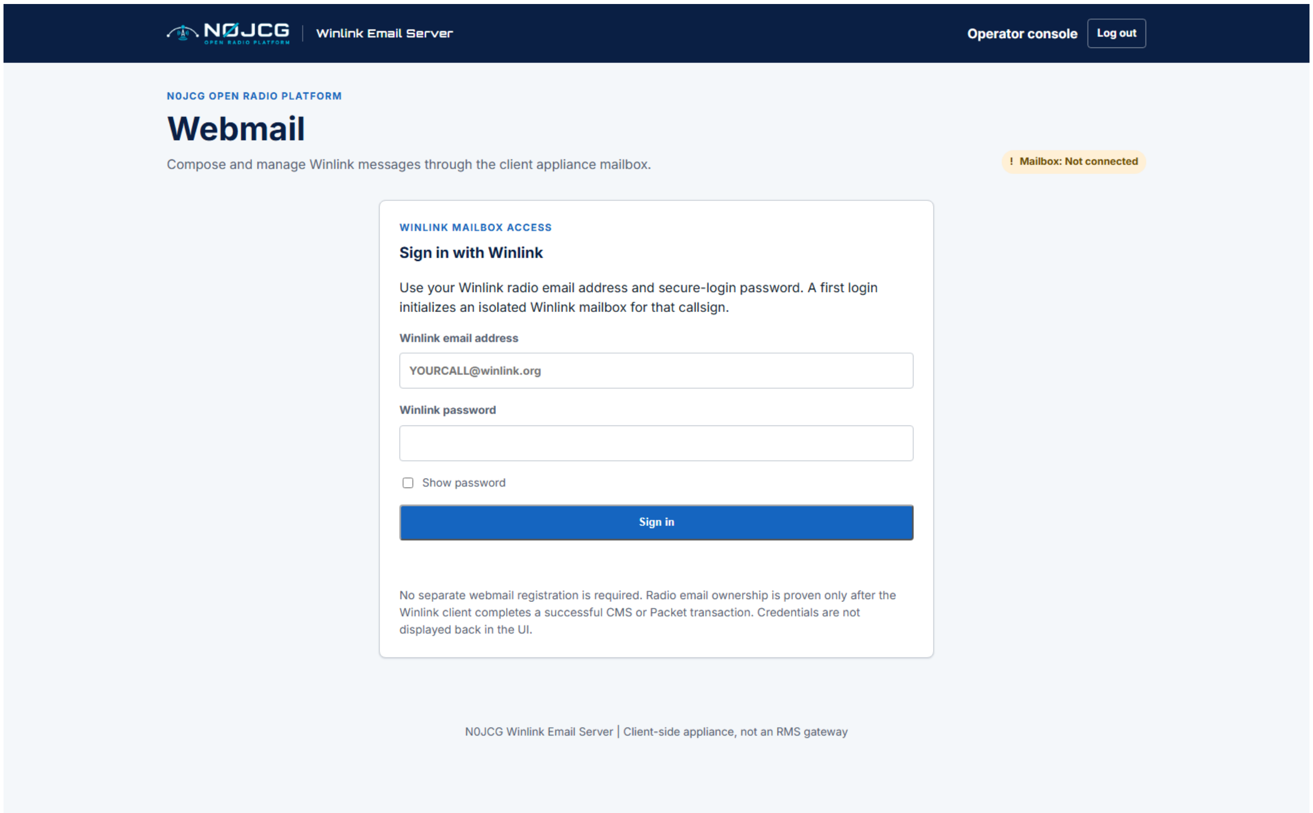

Figure 2. Current N0JCG Winlink webmail sign-in surface.

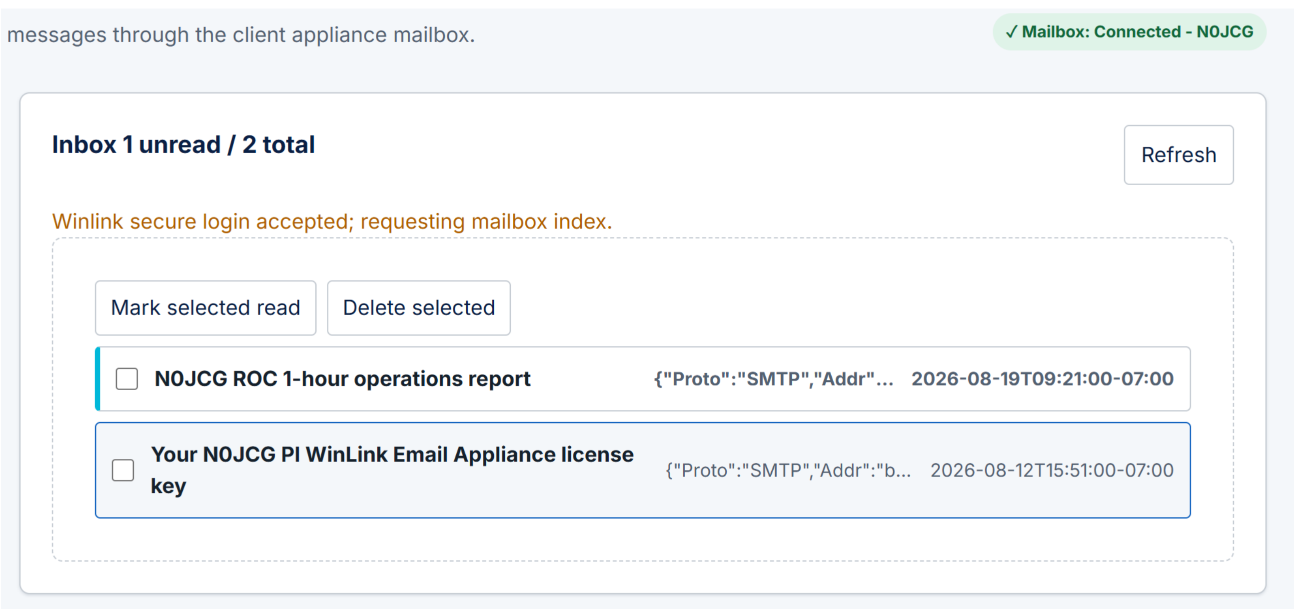

Figure 2. Webmail shows the authenticated mailbox and current synchronization state while a Packet RMS transfer is in progress.

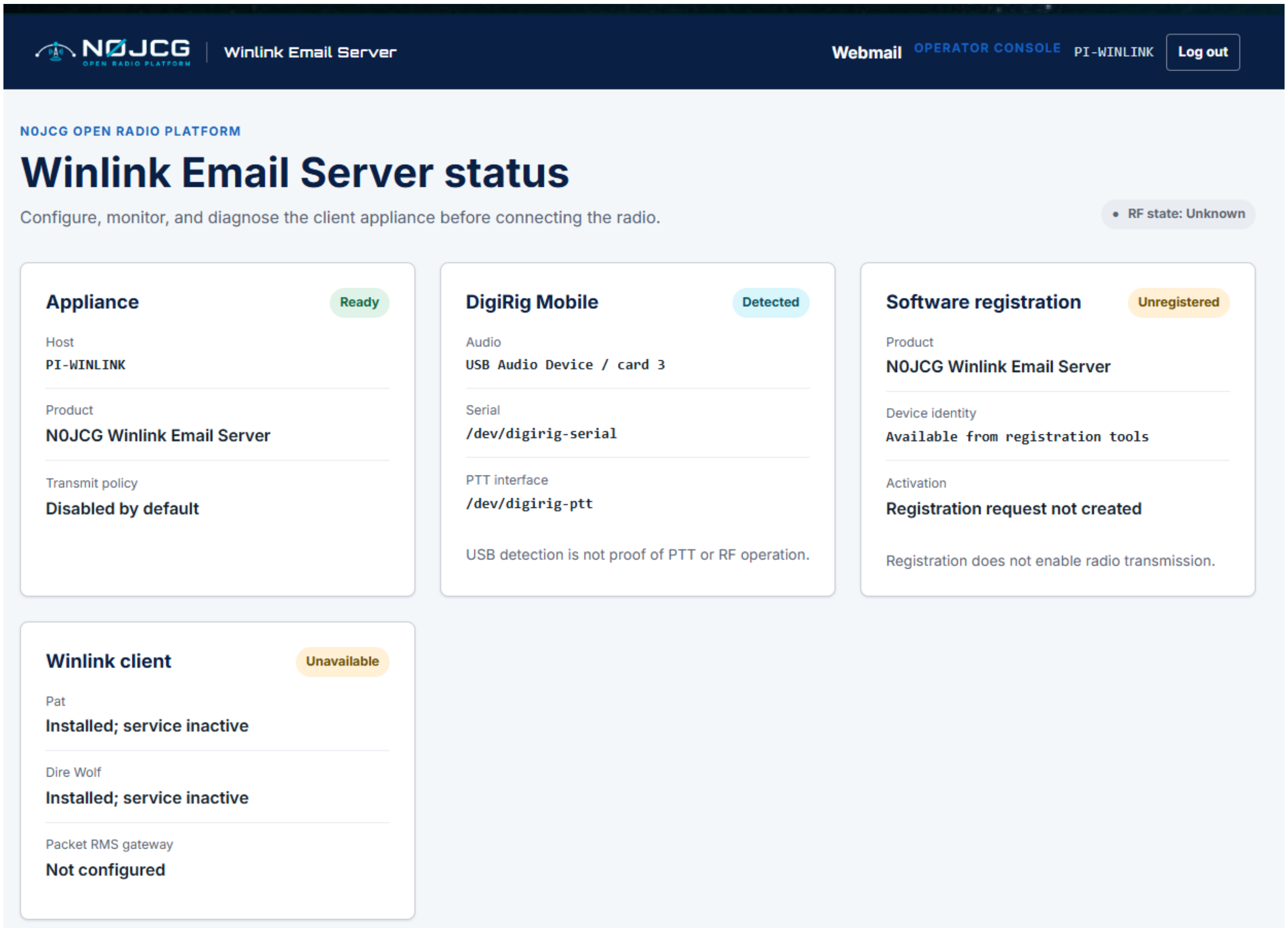

Figure 3. Operator status separates device detection, service readiness, RF state, and the transmit safety policy.

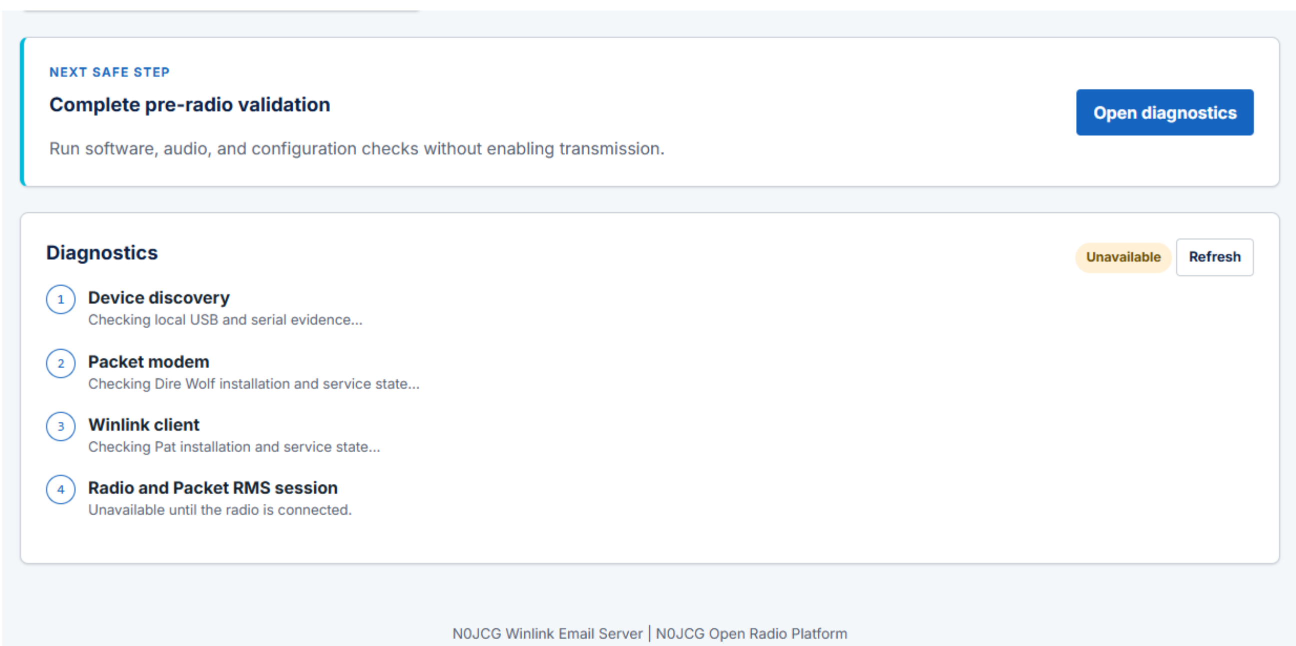

Figure 4. Operator diagnostics provides a safe pre-radio checklist and a clear next action.

4. Everyday webmail

Compose, templates, and drafts

Select Compose to open a blank form. At the top of the form, use Choose template to select an installed Winlink Standard Form, complete its fields, and insert an editable message. The Signature control is beside Choose template. Enter the Winlink recipient, subject, and message body. Select Save draft when work should remain local, or queue the message when the operator has enabled a verified transport path.

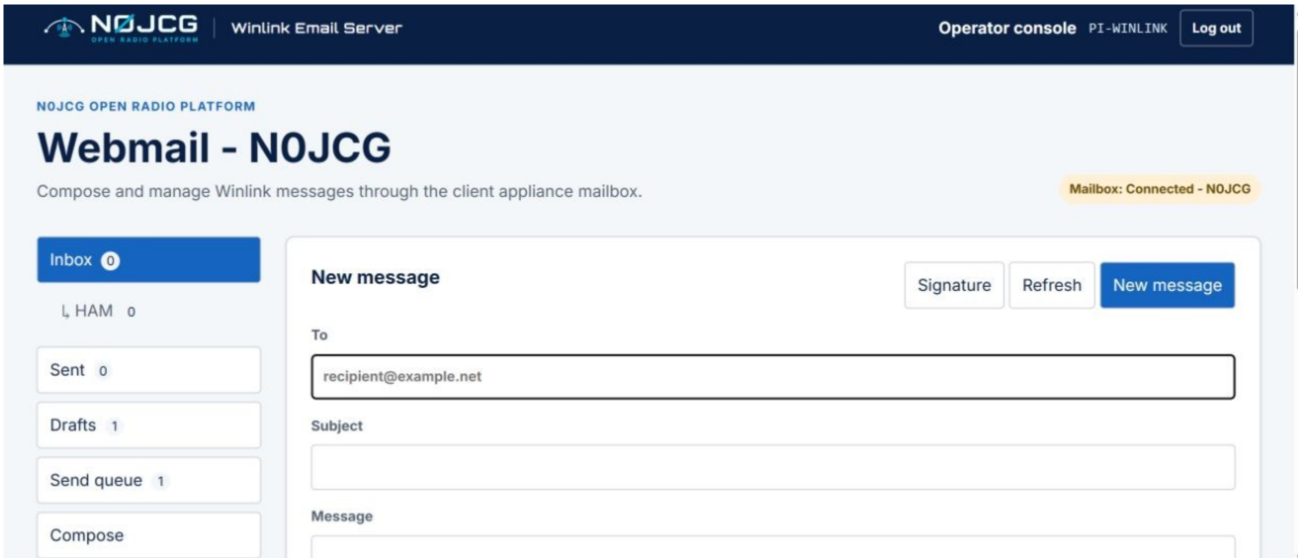

Figure 5. Compose panel with attachment controls and the N0JCG navigation shell.

Drafts, attachments, and signature

- The Drafts counter updates after saving. Open a draft to continue editing; queuing an opened draft removes it from Drafts and places it in Send queue.
- Attachments are limited to 100 KB. A warning appears above 10 KB because packet-radio transfer may be slow.
- Select Signature to save a per-user signature. It is restored after the next login for that Winlink callsign.
- Use Log out when leaving a shared workstation. Webmail uses a browser-session cookie by default; it does not remember users across browser sessions.

Folders

Custom folders are shown indented under Inbox. Create and delete folders from Manage folders, then drag an Inbox message onto a folder to organize it. Moving a message removes it from its original folder. Opening a message marks it read. Inbox, Sent, and custom folders provide checkboxes for bulk Mark selected read and Delete selected actions.

Send queue and automatic synchronization

Queue message stages the message into Pat’s official outbox and immediately starts a Packet RMS synchronization. The Send queue shows queued, staged, and transmission state. After a successful send, the message is removed from Send queue and its counter updates automatically. The operator configures automatic mailbox synchronization in the operator console under Packet station → Automatic mailbox sync (minutes), from 5 to 1440 minutes; the default is 30 minutes.

After Pat reports the final outgoing-transfer command, WES clears the transmitted queue item promptly while the remainder of the mailbox synchronization continues. This prevents a successful send from appearing stuck while the session closes.

Mailbox synchronization progress

After Winlink secure login is accepted, WES opens the webmail session and continues the mailbox transfer in the background. The status message advances through the RMS connection, mailbox index, proposal, message selection, download, and finalization stages. A message such as 0 new emails were received means the authenticated mailbox was checked and no new messages were available in that exchange. The Refresh control is disabled while a transfer is active so an operator cannot interrupt the current RF session.

If the radio session closes after one or more messages have been received, WES records the received messages and clears the stale in-progress state. A transport failure after authentication does not expose another user's mailbox and does not invalidate the logged-in callsign unless authentication itself has been lost.

5. Connect and verify the radio

1. Power off the radio and Raspberry Pi before changing audio/PTT cabling.
2. Connect the DigiRig Mobile to the radio with the correct radio-specific cables.
3. Connect the DigiRig Mobile directly to a Raspberry Pi USB 3 port or powered USB hub.
4. Keep the Pi 4 USB-C power/data port available for the USB Ethernet gadget; it is not a DigiRig host port.
5. Power the radio, DigiRig Mobile, and Raspberry Pi.
6. Confirm the operator console reports expected USB/audio/serial devices as Detected.
7. Configure the radio profile, audio levels, PTT method, and selected Packet RMS gateway.
8. Perform receive-only checks before enabling transmit behavior.
9. Enable transmission only after the operator has verified the radio path and local regulations.

SAFETY RULE

Do not enable transmit merely because a DigiRig or USB device is present. Detection is hardware evidence only; PTT and RF behavior require separate verification.

What the gateway means here

The Packet RMS gateway is an external Winlink service reached over the radio path. This product is the client-side email server and does not host or replace that gateway.

6. Troubleshooting

Mailbox unavailable

This means the Winlink account was authenticated, but the local mailbox service is not currently available. It is not a request for the user to type a command. Connect the radio and select Refresh to start a mailbox synchronization. If the message persists, the operator should review the client service and diagnostics.

Local delete versus Winlink Webmail

The WES Inbox, folders, and delete actions manage the local appliance mailbox. Winlink Webmail is a separate CMS view and may continue to show the same server-side message after WES downloads or locally deletes it. Remove a message separately through Winlink Webmail when server-side retention also needs to change.

A previous RF session is still active

Only one mailbox user and one RF session are allowed at a time. A new login terminates the previous session, releases the Pat/AGW connection, and allows the radio path to settle before starting again. After a failed exchange, wait for the visible failure state and the RF activity to stop before retrying. The operator diagnostics show the last client/RF event.

Login fails or remains on Validating through the Winlink client...

Confirm the Winlink address and secure-login password. Check the Pi network connection, radio power, DigiRig cabling, frequency, packet station ID, RMS target, and system time. The login screen intentionally remains open until secure login is accepted; no mailbox is exposed during this wait. A failed RF exchange returns the user to the login screen. Repeated failed attempts are rate-limited for protection.

Device detected but not ready

Check the correct radio cable, audio routing, serial/PTT configuration, and radio power. Do not enable transmit based on USB detection alone.

Deployment verification fails

Rerun the deployment helper from MSYS2 Bash and confirm the Pi IP address, SSH username, and SSH password. Use the diagnostic helper for a status report:

```
./deploy/inspect_pi_webmail.sh
```

Quick reference

Task	Location
Deploy or upgrade	./deploy/push_to_pi.sh in MSYS2 Bash
Diagnostics	./deploy/inspect_pi_webmail.sh
Operator console	http://PI-IP/ui/
User webmail	http://PI-IP/webmail/
Default SSH user	pi
Product host role	PI-WINLINK
Maximum attachment	100 KB
Attachment warning	Above 10 KB
Automatic mailbox sync	Operator setting, 5–1440 minutes; default 30
Login security	Webmail opens only after Winlink secure-login acceptance

SUPPORT CHECKLIST

When requesting help, include the appliance IP address, the page being used, the exact visible message, whether the radio is connected, and the output of inspect_pi_webmail.sh. Never include passwords.